AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M348
Applied Statistical Modelling

Final Examination
Spring Semester 2025/2026

Date: 10/June/2026

Number of Exam Pages: 7 (including this cover sheet)	**Time Allowed: 3 hours**

Instructions:

- Total Marks: **50**
- This exam consists of **4** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: Multiple choice Questions [10x1=10 marks]

For each of the following multiple-choice questions, select the most appropriate answer. Only one answer is correct for each question.

1. A small p-value in hypothesis testing indicates:
   a) Strong evidence against the null hypothesis
   b) Strong evidence in favor of the null hypothesis
   c) That the test statistic is small
   d) That the residuals are normally distributed
2. What is the purpose of a t-test in regression analysis?
   a) To test whether the overall model is significant
   b) To test whether an individual regression coefficient is significantly different from zero
   c) To test for multicollinearity
   d) To compare two models using deviance
3. In a regression model with categorical predictors, the intercept represents:
   a) The mean response for all categories combined
   b) The mean response for the reference category
   c) The slope of the first predictor
   d) The overall variance of the response
4. If regression lines for different groups are parallel, what does it indicate?
   a) There is no interaction effect between the predictors
   b) The model is non-linear
   c) The residuals are independent
   d) The model is overfitted
5. Logistic regression is a type of GLM used when:
   a) The response variable is continuous
   b) The response variable is binary
   c) The predictors are categorical
   d) The data are counts
6. What is the canonical link function for the binomial family in GLMs?
   a) Identity link
   b) Log link
   c) Logit link
   d) Square root link
7. Poisson regression is typically used for:
   a) Continuous outcomes
   b) Binary outcomes
   c) Count data
   d) Proportions
8. The null deviance in a GLM represents:
   a) The deviance of the model with all predictors included
   b) The deviance of a model including only the intercept

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c) The difference between predicted and observed values
d) The residual standard error

9. Transforming variables in regression can help:
   a) Satisfy model assumptions
   b) Increase multicollinearity
   c) Reduce sample size
   d) Decrease the number of predictors
10. Why might adding an interaction term improve a regression model?
   a) To allow the effect of one predictor to depend on the level of another predictor
   b) To reduce multicollinearity
   c) To remove outliers
   d) To simplify interpretation

PART 2: Parallel slopes models for the MTCARS dataset [13 marks]

ALL QUESTIONS ARE REQUIRED

In this exercise, we will analyze data from the mtcars dataset, which contains fuel consumption and other automobile design and performance characteristics for 32 car models. The focus is on modeling the relationship between a car's fuel efficiency and its weight, while accounting for the number of cylinders as a categorical factor.

The dataset includes the following variables relevant to this analysis:

- mpg: miles per gallon, representing fuel efficiency (continuous response variable)
- wt: weight of the car in thousands of pounds (continuous predictor)
- cyl: number of cylinders in the engine, treated here as a categorical variable with levels (4, 6, 8)

We fit a parallel slopes linear regression model, where the effect of weight on mpg is assumed the same across all cylinder groups, but the intercept may differ by cylinder category.

1) Fit a parallel slopes regression model with mpg as the response and wt and cyl as predictors. Store the fitted model in an object called model_parallel and display the summary.

The resulting output of part (1) are given below:

```
Residuals:
    Min      1Q  Median      3Q     Max
-4.5890 -1.2357 -0.5159  1.3845  5.7915
```

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Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  33.9908     1.8878  18.006  < 2e-16 ***
wt           -3.2056     0.7539  -4.252 0.000213 ***
cyl6         -4.2556     1.3861  -3.070 0.004718 **
cyl8         -6.0709     1.6523  -3.674 0.000999 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.557 on 28 degrees of freedom
Multiple R-squared:  0.8374,        Adjusted R-squared:  0.82
F-statistic: 48.08 on 3 and 28 DF,  p-value: 3.594e-11
```

2) Using the summary output, interpret the intercept (33.9908). What does it represent for a 4-cylinder car with weight = 0?
3) Interpret the coefficient for wt (-3.2056). What does this estimate imply about the effect of car weight on mpg?
4) Interpret the coefficients for cyl6 (-4.2556). How do this value affect the predicted mpg relative to 4-cylinder cars?
5) Based on the standard errors of the coefficients, which predictors are estimated with greater precision?
6) Explain the meaning of the t-values. How do they relate to the statistical significance of each predictor?
7) Using the Pr(>|t|) column, which predictors are statistically significant at the 0.05 level?
8) How do the Multiple R-squared (0.8374) and Adjusted R-squared (0.82) values inform you about the proportion of variation explained by the model? Why the adjusted R-squared is slightly lower?
9) Interpret the F-statistic (48.08) and its p-value. What does this tell you about the overall model?
10) Why this model is called a parallel slopes model? How does this assumption affect the interpretation of the wt coefficient across cylinder groups?
11) If you wanted to improve the model, what potential modifications could you consider? In your answer, explain:
    - When and why you might add interaction terms (e.g., to allow the effect of weight on mpg to differ across cylinder groups).
    - When and why you might transform variables (e.g., to address non-linearity, heteroscedasticity, or improve model).

PART 3: Log-linear models of two-way contingency table data [13 marks]

ALL QUESTIONS ARE REQUIRED

In this part, we will fit log-linear models of two-way contingency table data. For that, we will use a dataset describing the relationship between smoking status and lung disease. The data are presented as a two-way contingency table with:

• Two rows for the factor smoking status (one for the level smoker and one for non-smoker)

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• Two columns for the factor lung disease (one for the level yes and the other for no)

• The counts of individuals from the sample (of 800 individuals) in each of the four cells of the table

1. Fit a log-linear model taking count to be the response, and taking smoking status and lung disease as explanatory variables. Store the results of the fitted model in the object modelFit, and display a summary of the results.

```
Deviance Residuals:
    1      2      3      4
 3.007  -1.732  -3.368   1.677

Coefficients:
             Estimate  Std. Error  z value  Pr(>|z|)
(Intercept)    5.737e+00  5.351e-02  107.21   <2e-16 ***
smokingSmoker -2.151e-14  7.071e-02    0.00        1
diseaseYes    -1.237e+00  8.467e-02  -14.61   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)
```

```
Null deviance: 282.177  on 3  degrees of freedom
Residual deviance:  26.203  on 1  degrees of freedom
AIC: 59.906

Number of Fisher Scoring iterations: 4
```

2. What type of model was fitted?
   a. Linear regression
   b. Logistic regression
   c. Poisson log-linear model
   d. ANOVA

3. Which of the following best describes the interpretation of the intercept?
   a. The expected log count for smokers with lung disease
   b. The expected log count for the reference group (non-smokers with no lung disease)
   c. The total number of individuals in the sample
   d. The difference in counts between smokers and non-smokers

4. Which factor combination is the reference category in the fitted model?
   a. Smoker, Yes
   b. Smoker, No
   c. NonSmoker, No
   d. NonSmoker, Yes

5. What does a significant coefficient for diseaseYes mean?
   a. Lung disease status affects the expected counts in the contingency table
   b. Smoking and lung disease are independent
   c. The counts are normally distributed
   d. The model has perfect fit
6. Which statement best describes the meaning of the residual deviance being 26.203 on 1 degree of freedom?

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a. The model fits poorly
b. The model fits well since residual deviance is small relative to df
c. The model is overfitted
d. The model has zero residual error

7. The estimated mean count for the reference category (non-smokers with no lung disease) is approximately:
   a. e^5.737 ≈ 309
   b. e^-1.237 ≈ 0.29
   c. e^(5.737−1.237) ≈ 90
   d. e^(5.737+1.237) ≈ 900

8. Write down the fitted log-linear model equation using the output coefficients.
9. Based on the fitted model, what is the expected count for smokers without lung disease?
10. Interpret the coefficient for diseaseYes.
11. What assumption about the response variable justifies using a Poisson model in this context?
12. How would you interpret a large reduction between the null deviance and the residual deviance?
13. How could the fitted model be used to estimate expected counts for each cell of the contingency table?

PART 4: Fitting a Binomial Generalized Linear Model for OU students dataset [14 marks]

ALL QUESTIONS ARE REQUIRED

In this exercise, you will fit a binomial generalized linear model (GLM) with a logit link function using data from the OU student's dataset. The goal is to model the relationship between students' final exam performance and their previous academic performance and age.

The dataset contains the following variables:

- examScore: the final exam score, taking integer values from 0 to 100
- bestPrevModScore: the best previous final module score, taking values from 0 to 100
- age: the age of the student in years (with a small random adjustment of −2, −1, 0, 1, or 2 added)

The response variable examScore represents the number of marks obtained out of 100 possible marks

```
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-15.5478  -2.1775   0.6821   2.8365  11.5252
Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)     -3.3386601  0.0537657 -62.096   <2e-16 ***
bestPrevModScore  0.0471885  0.0006050  78.001   <2e-16 ***
age             -0.0039868  0.0004384  -9.094   <2e-16 ***
---
```

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```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 31830  on 1795  degrees of freedom
Residual deviance: 25381  on 1793  degrees of freedom
AIC: 33687

Number of Fisher Scoring iterations: 4
```

1. Explain why the response variable examScore can be modelled using a binomial distribution.
2. Explain the difference between a Bernoulli GLM and a binomial GLM.
3. Why does R require the response data for a binomial GLM to be stored in two columns?
4. What link function is used by default when the argument family = "binomial" is specified in the glm() function?
5. Write the R code to fit a generalized linear model with response variable examScoreMatrix and explanatory variables bestPrevModScore and age, assuming a binomial distribution. Store the fitted model in an object called modelFit.
6. Write the mathematical form of the fitted model using the logit link function.
7. Interpret the meaning of the coefficient associated with bestPrevModScore.
8. What does the intercept represent in this model?
9. What statistical test is used in the model summary to determine whether each coefficient is significantly different from zero?
10. What information do the standard errors of the coefficients provide?
11. For a new student with bestPrevModScore = 79.2 and age = 64, how would you calculate the predicted probability of success using the fitted model?
12. Explain the role of the logit link function in a binomial GLM and write down its mathematical expression relating the success probability pi to the linear predictor ηi.
13. Using the notation from the model, describe how the fitted mean response μi is calculated from the number of trials Ni and the fitted success probability pi.

End of questions

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